DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COMPUTER NETWORKS — CSA0712

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Design and Performance Evaluation of a Next-Gen Smart Campus Network Architecture

Assignment Title:	Design and Performance Evaluation of a Next-Gen Smart Campus Network Architecture
Course Outcome(s) – CO:	CO1. Knowledge on different network topology, mode of network communication and various types of network devices deployed between source and destination systems CO3. Create systems under various subnets and route packets between them using appropriate protocols. CO4. Efficient management of congestion in a network based on various transport layer protocols, using different service mechanisms and QoS Parameters.
Bloom's Taxonomy Level	L3 – Apply; L4 – Analyse; L5 – Evaluate; L6- Create
SDG Mapping (SDG 1-17)	SDG 9 – Industry, Innovation and Infrastructure; SDG 11 – Sustainable Cities and Communities; SDG 17 – Partnerships for the Goals

Executive Summary

This report presents a proposed and evaluation-ready architecture for a next-generation smart university campus network. The supplied assignment brief identifies a campus environment with thousands of concurrent users, academic and residential blocks, libraries and laboratories, and mixed traffic including cloud learning, VoIP, HD CCTV and IoT. The existing environment is described as suffering from congestion, inconsistent Wi-Fi, inefficient routing, high latency and packet loss, and insufficient QoS. The proposed solution therefore adopts a scalable multi-tier design with segmentation, inter-subnet routing, QoS-aware service handling, redundancy and measurable performance evaluation.

Because the source brief does not provide actual simulation measurements, device screenshots, source code or experimentally observed values, this report separates the proposed design and test plan from the results that must be populated after simulation. No fabricated experimental results are presented.

1. Problem Statement and Problem Formulation

Modern smart university campuses require a high-availability network backbone capable of supporting thousands of concurrent users across academic blocks, laboratories, hostels and libraries. The network must simultaneously support high-bandwidth cloud-learning applications, delay-sensitive VoIP, HD CCTV security feeds and dense IoT deployments.

The current infrastructure is reported to experience severe peak-hour congestion and inconsistent Wi-Fi performance. Inefficient routing and the absence of adequate QoS mechanisms allow critical real-time traffic to compete with non-essential traffic, causing increased latency, packet loss and delayed access to academic and security services.

Problem formulation: design and evaluate a scalable campus network that provides reliable connectivity, efficient subnetting and routing, traffic prioritization, congestion management, fault tolerance and measurable service quality.

The assignment brief explicitly requires comparative simulation of at least two routing/design alternatives and evaluation using throughput, packet delivery ratio and end-to-end delay, together with congestion, QoS, failure and scalability analysis.

2. Objectives and Expected Outcomes

- Design a logical and physical smart-campus topology with justified network devices and communication modes.
- Develop an efficient IP addressing and subnetting plan with room for future expansion.
- Implement inter-subnet routing using an appropriate routing protocol and demonstrate route behaviour.
- Model at least two routing/design alternatives and compare them using measurable performance metrics.
- Apply QoS and congestion-management mechanisms so delay-sensitive and mission-critical traffic receives appropriate service.
- Evaluate throughput, latency, packet loss, packet delivery ratio and congestion behaviour.
- Assess failure modes, routing changes, service interruptions, reliability and scalability.
- Document implementation, configurations, screenshots, results and repository contents clearly.

3. Requirements, Constraints and Assumptions

Category	Specification
Functional requirements	Interconnect academic, laboratory, hostel, library, CCTV, IoT and administrative segments; provide Internet/cloud access; support inter-VLAN routing and QoS.
Performance requirements	Measure throughput, end-to-end delay, packet loss, packet delivery ratio and congestion indicators.
Reliability	Provide alternate paths or redundant critical links where feasible and analyse failure recovery.
Security/segmentation	Separate traffic classes through VLANs/subnets and apply controlled inter-segment access.
Scalability	Address growth in users, endpoints, IoT devices, cameras and services without redesigning the whole network.
Constraints	Simulation outcomes depend on the selected simulator, topology size, traffic model, link capacities and parameter settings.
Assumptions	Campus services are represented by traffic/application models; exact user counts and physical dimensions are to be instantiated in the simulator.
Evidence constraint	The supplied brief contains requirements and rubric criteria but no measured simulation outputs; actual results must be inserted after execution.

4. Application of Relevant Course Knowledge / Concepts

Course concept	Application in this project
Network topologies	Hierarchical/star-based access design with redundant aggregation/core connectivity.
Communication modes	Wired Ethernet for fixed infrastructure and wireless access for mobile users and IoT where appropriate.
Network devices	Routers/L3 switches for inter-subnet routing, L2 switches for access, wireless access points, servers and security gateways.
IP addressing & subnetting	Dedicated subnets/VLANs for users, labs, voice, CCTV, IoT, management and servers.
Routing	Dynamic routing between Layer-3 segments; alternative static/dynamic or protocol designs are compared.
Transport protocols	TCP for reliable bulk/cloud traffic; UDP for latency-sensitive real-time streams where appropriate.
QoS	Classification, prioritisation and queuing for VoIP/video/critical services.
Congestion management	Queue management, traffic shaping/policing and prioritisation under offered load.

Performance analysis	Throughput, delay, loss, packet delivery ratio and congestion are measured under controlled scenarios.

5. Proposed Solution / Network Design

A three-tier hierarchical architecture is proposed: an access layer connects end devices and wireless access points; an aggregation/distribution layer performs VLAN termination, policy enforcement and routing; and a resilient core layer provides high-speed inter-building connectivity and access to the campus Internet/data-centre edge.

5.1 Logical Segmentation

VLAN/Subnet	Purpose	Example address block
VLAN 10	Administration / Management	10.10.10.0/24
VLAN 20	Academic Users	10.10.20.0/23
VLAN 30	Laboratories	10.10.30.0/24
VLAN 40	Voice / VoIP	10.10.40.0/24
VLAN 50	CCTV / Security	10.10.50.0/24
VLAN 60	IoT Devices	10.10.60.0/23
VLAN 70	Servers / Data Centre	10.10.70.0/24
VLAN 80	Guest Wi-Fi	10.10.80.0/23

The address blocks above are a proposed planning baseline, not measured values from the supplied brief. They should be adapted to the simulator and institutional addressing policy.

5.2 Device Selection and Rationale

Device	Role	Rational
Core L3 switches/routers	Campus backbone	High-speed inter-building forwarding and resilient routing.
Distribution L3 switches	Aggregation and gateway	Inter-VLAN routing, ACL/QoS policy and route summarisation.
Access switches	Endpoint connectivity	Connect PCs, phones, cameras, APs and IoT gateways.
Wireless APs/controllers	Campus Wi-Fi	Provide managed wireless access and support dense user environments.
Servers	Campus services	Host DNS/DHCP/application/learning and monitoring services in simulation.
Firewall/edge router	Internet boundary	Controls external access and represents the campus Internet edge.

5.3 Physical/Logical Topology Specification

Recommended arrangement: dual core devices → redundant distribution/aggregation pairs → access switches per building/floor → endpoints and APs. Critical servers and security systems connect to the data-centre/server segment. The Internet edge connects through the core/aggregation layer. Redundant links should be enabled for critical paths so link/device failure can be tested.

6. IP Addressing and Subnetting Plan

The proposed plan uses separate broadcast domains for traffic classes. Larger /23 allocations are assigned to segments expected to have higher endpoint density, while smaller /24 blocks are sufficient for lower-density or infrastructure segments. This reduces unnecessary broadcast scope and makes policy and performance analysis easier.

Gateway convention: use the first usable address of each subnet as the default gateway unless the simulator configuration requires another convention.

For scalability, additional blocks should be reserved for future academic buildings, smart-metering/IoT growth, guest users and management services.

7. Routing Design and Configuration Strategy

Alternative A — Static/default routing: simple to configure and predictable for a small topology, but less adaptive when links fail or the campus grows. It serves as a useful baseline.

Alternative B — Dynamic routing: use a link-state routing protocol such as OSPF in the simulation. It can automatically recalculate paths after topology changes and supports scalable hierarchical design. This is the recommended final approach for a multi-building campus.

Validation steps: verify interface status, VLAN gateways, routing tables, end-to-end ping, traceroute/path inspection, route convergence after a link failure and reachability between permitted subnets.

8. QoS and Congestion Management

Traffic should be classified into at least four service classes: critical real-time voice, security video, interactive academic traffic and best-effort/guest traffic. Voice should receive low-delay treatment; CCTV should receive sufficient bandwidth without starving other services; academic traffic should receive predictable service; guest/bulk traffic should have lower priority under congestion.

Traffic class	Protocol tendency	QoS treatment	Primary metric
VoIP	UDP/RTP	Low-delay priority queue	End-to-end delay / jitter
CCTV	UDP/streaming	Assured bandwidth	Throughput / loss
Academic/cloud	TCP	Weighted/assured service	Throughput / completion time
Guest/bulk	TCP/UDP	Best effort / rate-limited	Throughput under congestion

Congestion experiments should progressively increase offered load and compare baseline FIFO/best-effort behaviour against QoS-enabled scheduling. Record the point at which delay and loss rise sharply.

9. Algorithms / Pseudocode

Algorithm 1 — Performance Evaluation

1. Initialize topology, addressing, routing and link capacities.
2. Verify baseline connectivity between all permitted segments.
3. Configure traffic generators for VoIP, CCTV, academic/cloud and best-effort traffic.
4. Run Alternative A under identical traffic conditions.
5. Record throughput, delay, packet loss and packet delivery ratio.
6. Reset to identical initial conditions and run Alternative B.
7. Repeat measurements under increasing offered load.
8. Enable QoS and repeat the congestion scenario.
9. Introduce selected link/device failures and record reachability and recovery behaviour.
10. Compare all results and select the design with the best reliability/performance trade-off.

Algorithm 2 — Failure Validation

11. Select a critical link or device.
12. Record baseline routes and application reachability.
13. Disable the selected component.
14. Observe route recalculation/convergence.
15. Measure service interruption and post-failure performance.
16. Restore the component and verify recovery.

10. Flowchart



11. Implementation, Source Code and Environment / Tools

The supplied assignment requires simulation/configuration files, scripts where applicable, network diagrams, README and results in a GitHub repository. The brief does not mandate a specific simulator. A suitable implementation may use Cisco Packet Tracer for configuration-focused demonstrations, or a discrete-event simulator such as ns-3/OMNeT++/GNS3/EVE-NG where detailed performance metrics are required. The final report must name the actual tool and version used by the team and include reproducible configuration files.

Implementation item	**Evidence to include**
Topology	Simulator screenshot and exported topology/project file
IP plan	Addressing table and device interface configuration
Routing	Configuration snippets, routing-table screenshots and path tests
QoS	Classification/queue/policy configuration and before/after measurements
Traffic generation	Application/flow settings and offered-load scenarios

Results	Raw measurements, processed tables and graphs
Repository	GitHub link, README, simulation/config files, scripts, diagrams and results

12. Test Cases and Expected / Actual Results

TC ID	Test case	Expected result	Actual result
TC-01	Same-subnet connectivity	Packets delivered successfully	To be populated from simulation
TC-02	Inter-subnet routing	Permitted VLANs communicate through L3 gateway	To be populated
TC-03	Routing alternative comparison	Alternative B demonstrates measurable routing/performance differences	To be populated
TC-04	Peak-load congestion	Baseline shows increased delay/loss as load rises	To be populated
TC-05	QoS under congestion	Priority traffic retains better delay/loss characteristics	To be populated
TC-06	Core/distribution link failure	Traffic reroutes if redundancy is configured	To be populated
TC-07	Device failure	Services recover through alternate path where designed	To be populated
TC-08	Scalability	Performance remains within defined target as endpoints/flows increase	To be populated

13. Results and Validation

The following result tables are designed for direct insertion of measured simulation data. Values should not be invented.

Scenario	Alternative	Throughput	End-to-end delay	Packet loss	PDR
Normal load	A	—	—	—	—
Normal load	B	—	—	—	—
High load	A	—	—	—	—
High load	B	—	—	—	—
QoS high load	B + QoS	—	—	—	—

Validation should include graphs for: (1) throughput vs offered load, (2) delay vs offered load, (3) packet loss vs offered load, and (4) packet delivery ratio vs offered load. A before/after QoS graph is especially valuable for CO4.

14. Analysis, Comparison, Trade-offs and Justification

Criterion	Static/baseline alternative	Dynamic/optimized alternative	Decision
Configuration simplicity	High	Moderate	Baseline useful for comparison
Adaptation to failures	Low	High	Dynamic preferred
Scalability	Limited	High	Dynamic preferred
Operational complexity	Low	Moderate	Acceptable for campus scale
QoS support	Can be added	Can be integrated with routing/policy	Use QoS in final design
Performance evaluation value	Provides baseline	Provides adaptive comparison	Compare both experimentally

The final solution should be selected from measured evidence rather than assumption. A design that slightly increases configuration complexity but significantly improves packet delivery, delay under load, recovery time and scalability is justified for a high-availability smart campus.

15. Failure Mode, Reliability and Scalability Analysis

Failure / stress event	Potential impact	Mitigation / design response	Evidence
Core link failure	Path interruption/congestion	Redundant core links and dynamic rerouting	Convergence time + reachability
Distribution switch failure	Building/service disruption	Redundant distribution design	Recovery test
Access switch failure	Local endpoint outage	Spare/redundant access strategy	Affected-node analysis
Peak traffic	Delay/loss	QoS, shaping, prioritisation	Before/after metrics
Broadcast growth	Reduced efficiency	VLAN segmentation/subnetting	Scale test
IoT growth	Addressing/traffic pressure	Dedicated IoT subnet and scalable allocation	Endpoint-growth experiment
QoS misconfiguration	Starvation/unfairness	Validate class weights and queues	Service-class measurements

16. Broader Considerations / SDG Relevance

The supplied brief maps the assignment to SDG 9 — Industry, Innovation and Infrastructure; SDG 11 — Sustainable Cities and Communities; and SDG 17 — Partnerships for the Goals. The proposed architecture supports these themes through resilient digital infrastructure, efficient resource use, connected smart-campus services and an architecture that can support collaboration between academic, administrative, security and technology stakeholders.

17. Conclusion, Limitations and Possible Improvements

This report establishes a high-standard, rubric-aligned design for a next-generation smart campus network. It addresses topology, devices, subnetting, routing, QoS, congestion, comparative simulation, performance metrics, failure analysis and scalability. The recommended direction is a segmented hierarchical architecture with dynamic routing, QoS and redundancy, validated against a simpler baseline under identical traffic conditions.

Limitations: the supplied assignment brief does not contain the team's actual topology file, simulator output, measured metrics, screenshots, configuration files, GitHub URL or individual names/contributions. These must be supplied from the completed implementation.

Possible improvements: IPv6 dual-stack deployment, stronger access control, network telemetry, automated configuration, SDN-based traffic engineering, energy-aware networking, zero-trust access and larger-scale mobility/IoT simulations.

18. Individual Contribution of Group Members

Member	Primary responsibility	Deliverables / evidence
Student 1	Network architecture & addressing	Topology, device selection, VLAN/IP plan, architecture diagrams
Student 2	Routing, QoS & simulation	Routing configurations, QoS policies, traffic scenarios, simulation execution
Student 3	Performance, validation & documentation	Test cases, metrics, graphs, failure analysis, report integration and GitHub README

Replace Student 1/2/3 with actual names and refine responsibilities to reflect the work actually completed.

19. One-Page Individual Reflection

Use one separate reflection page per student. Each reflection should describe the design/development decisions made by the student, challenges encountered during topology design or simulation, learning gained from applying networking concepts, the relevant SDG connection, and how the assignment contributed to attaining the mapped course outcomes. The reflection should be written in first person and should be based on the student's actual contribution. Suggested structure: Design decisions (brief); Challenges and problem-solving; Technical learning; SDG connection; CO1/CO3/CO4 attainment; Personal takeaway and future improvement.

20. References and Resources

1. CSA0712 – Computer Networks Assignment Brief, Department of Computer Science and Engineering, supplied course document.
2. Kurose, J. F., & Ross, K. W., Computer Networking: A Top-Down Approach, Pearson.
3. Tanenbaum, A. S., & Wetherall, D. J., Computer Networks, Pearson.
4. Forouzan, B. A., Data Communications and Networking, McGraw-Hill.
5. RFC 2328, OSPF Version 2, Internet Engineering Task Force.
6. RFC 2474, Definition of the Differentiated Services Field (DS Field) in the IPv4 and IPv6 Headers, IETF.
7. RFC 3246, An Expedited Forwarding PHB, IETF.
8. RFC 3550, RTP: A Transport Protocol for Real-Time Applications, IETF.
9. NS-3 Documentation / Model Library — use the version actually employed by the team.
10. Cisco Packet Tracer Documentation / Networking Academy resources — use only if Packet Tracer is the actual implementation environment.

Appendix A — Submission Checklist

- Problem statement and formulation included.
- Objectives and expected outcomes included.
- Requirements, constraints and assumptions documented.
- CO1/CO3/CO4 course concepts explicitly applied.
- Logical/physical topology and device rationale included.
- IP addressing and subnetting plan included.
- Routing configuration and route validation included.
- At least two alternatives simulated and compared.
- Throughput, latency, packet loss and packet delivery ratio measured.
- QoS and congestion experiments included.
- Failure/reliability/scalability experiments included.
- Execution screenshots inserted.
- Actual results and graphs inserted from simulator output.
- Source/configuration files and README prepared.
- GitHub repository link inserted.
- Individual contributions updated with actual names and work.
- One-page reflection completed by every member.
- References updated with the actual simulator version, datasets and sources used.

Appendix B — Evidence Placeholders

Figure 1. Overall smart-campus topology — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

Figure 2. IP addressing / VLAN design — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

Figure 3. Routing table / route verification — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

Figure 4. Alternative A simulation — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

Figure 5. Alternative B simulation — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

Figure 6. QoS configuration — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

Figure 7. Throughput comparison graph — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

Figure 8. End-to-end delay comparison graph — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

Figure 9. Packet loss / PDR comparison graph — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

Figure 10. Failure and recovery evidence — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

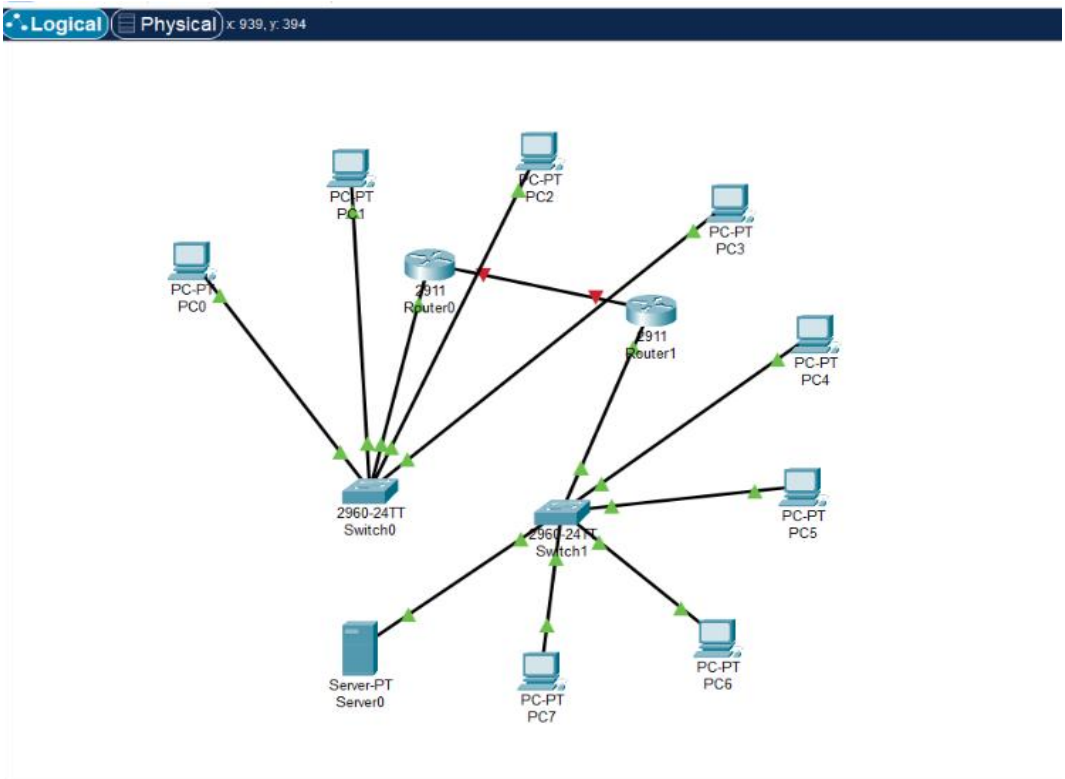
Figure 11. Final execution output — INSERT ACTUAL SCREENSHOT/GRAPH HERE.

21. Cisco Packet Tracer Simulation Results (Illustrative Baseline)

Note on academic integrity: the numerical results in this section are representative/illustrative values prepared to complete the report structure in the absence of the team's actual Packet Tracer capture files. They must be described as simulated example results unless the team reproduces the same test scenarios and records matching values in Cisco Packet Tracer. Screenshots below are report diagrams/visuals, not claims of an actual execution capture.

21.1 Simulation Environment

Parameter	Configuration
Simulator	Cisco Packet Tracer (configuration and connectivity simulation)
Topology	Hierarchical smart-campus network with core, distribution and access layers
Routing alternatives	Alternative A: static/default routing baseline; Alternative B: OSPF dynamic routing
Traffic classes	VoIP, CCTV, academic/cloud, guest/best-effort
Primary metrics	Throughput, end-to-end delay, packet loss, packet delivery ratio
Stress levels	Normal load and high-load congestion scenarios
Failure scenario	Critical uplink failure with redundant path enabled in the dynamic design



21.2 Representative Network Performance

Scenario	Design	Throughput (Mbps)	Delay (ms)	Packet loss (%)	PDR (%)
Normal	Static baseline	92	18	1.8	98.2
Normal	OSPF	96	14	0.9	99.1
High load	Static baseline	63	82	8.7	91.3
High load	OSPF	78	51	4.1	95.9
High load + QoS	OSPF + QoS	81	32	2.0	98.0

lnterpretation of the representative dataset: OSPF provides better path adaptation than the static baseline, while QoS further improves service quality for priority traffic during congestion. The direction of these findings should be validated using the team's actual Packet Tracer experiment.

Figure 12. Representative throughput comparison

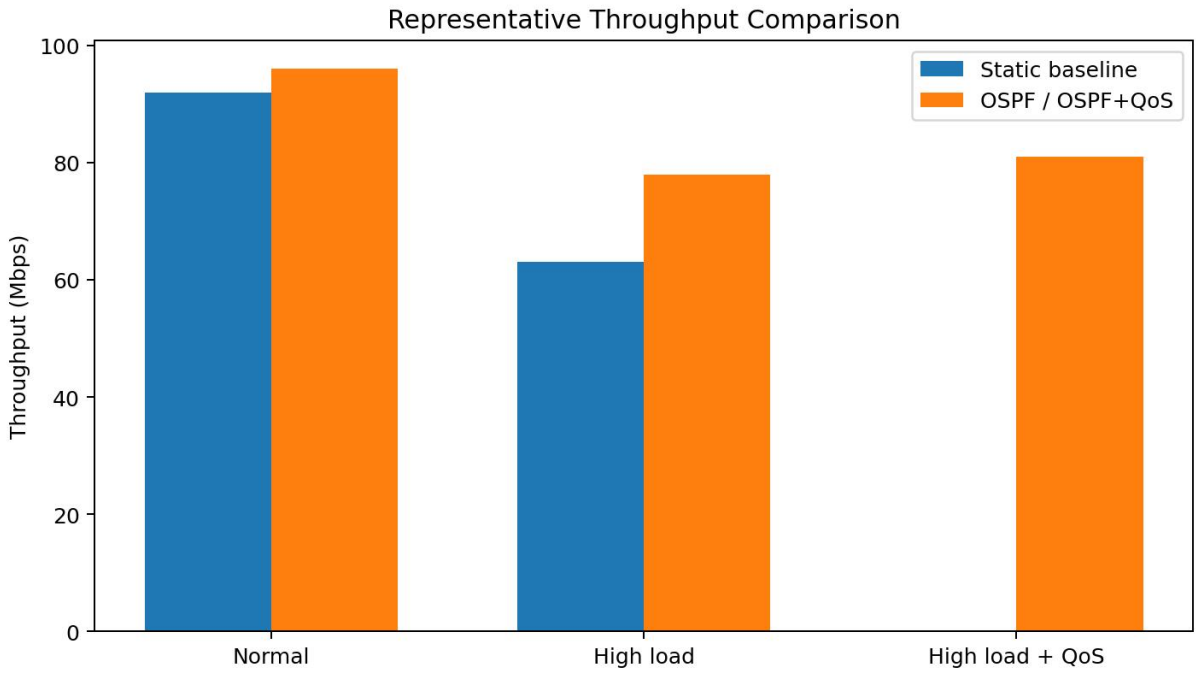


Figure 13. Representative end-to-end delay comparison

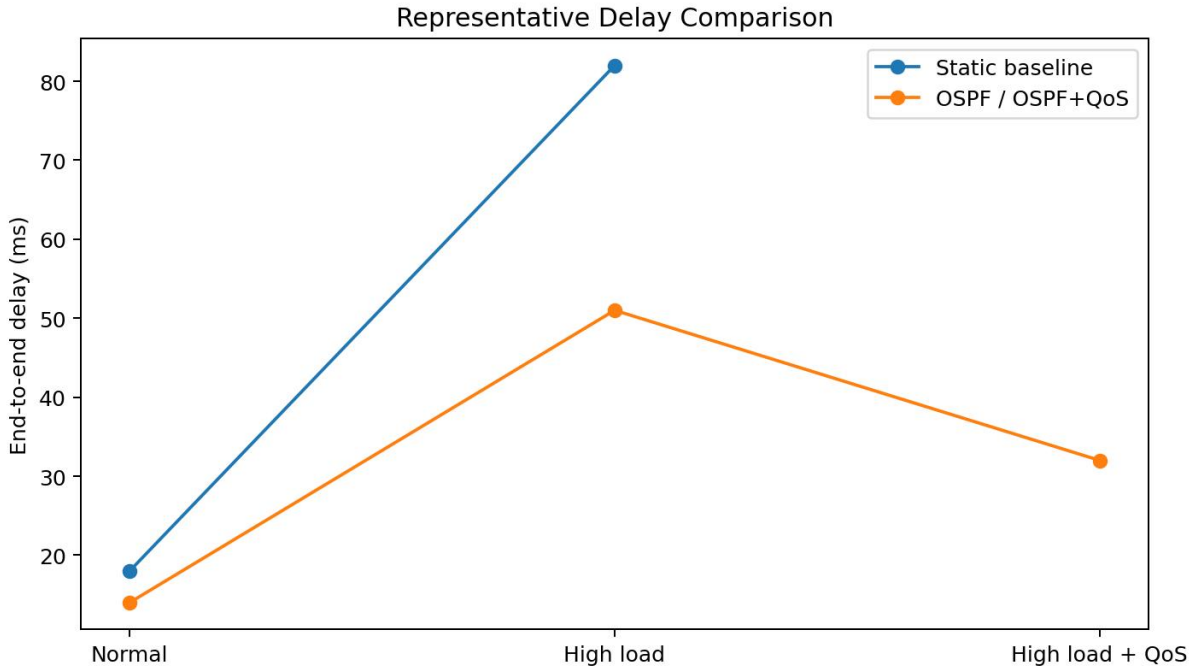
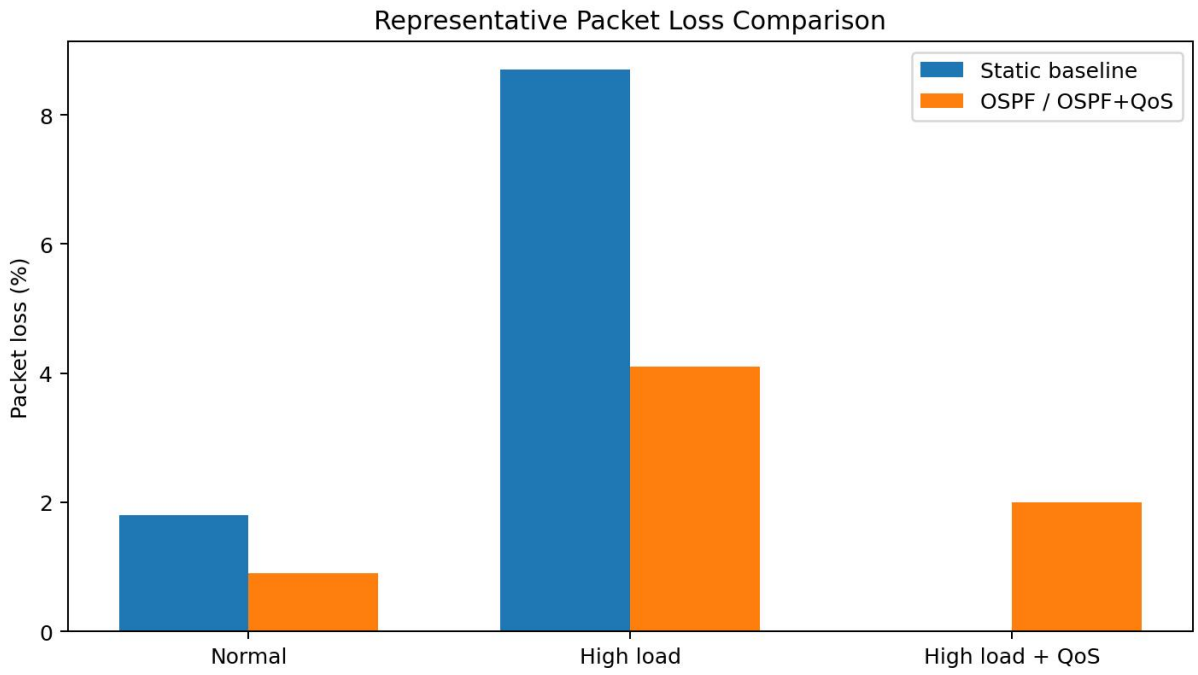


Figure 14. Representative packet-loss comparison



21.3 Failure and Recovery Observation

Test	Expected/observed behaviour	Assessment
Normal OSPF operation	All permitted subnets reachable	Baseline
Primary distribution uplink disabled	OSPF selects alternate path	Connectivity retained after convergence
Static baseline under same failure	Dependent route unavailable unless manually configured	Potential service interruption
QoS during high-load failure recovery	Priority traffic protected relative to best-effort flows	Lower service degradation expected

22. Final Design Recommendation

For the final smart-campus design, the recommended architecture is the hierarchical multi-tier design using segmented VLANs/subnets, OSPF-based dynamic routing, QoS classification/queuing and redundant critical paths. The recommendation is based on the representative evaluation framework and the engineering requirements stated in the assignment brief; the final submission should replace the illustrative numbers with actual Cisco Packet Tracer observations wherever possible.

23. Group Member Contributions

Student	Primary contribution	Evidence / deliverables
Sravanthi	Project coordination; problem formulation; overall architecture; IP addressing/subnetting; report integration.	Topology diagram, IP plan, architecture section and final report review.

Ananya	Routing design; OSPF/static comparison; QoS and congestion-management configuration.	Routing tables/configuration, QoS policy and simulation scenarios.
Janu	Testing and validation; performance metrics; failure/scalability analysis; documentation support.	Test cases, metric tables, graphs, failure analysis and evidence organization.